

Class: CMPM 151 - Algorithmic Music for Games

Professor: Dr. Brian Hansen

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Group Members: Andrew Degan, Ben Nguyen

Final Project Writeup

Audio Critique:

What is the purpose/function/genre of the game scene, and how does the audio complement it?

Our game scene is a bright, cheerful, exploration-based platformer that deliberately evokes feelings of the original NES Super Mario Bros., a tuneful, light-hearted world built around movement and discovery. The audio is designed to complement that genre by reinforcing its playful energy and its sense of an inviting, living space. Rather than a single looping track, the music is generative and non-repeating, which supports the relaxed, open-ended mood of exploration. A day/night cycle drives one of the scene's defining audio behaviors: as day turns to night, the music switches from the higher melody voice to a deeper-toned voice and the birds stop chirping in the background, reinforcing the passage of time through sound rather than relying on visuals alone. This mirrors the scene itself, which is colorful and inviting during the day but grows moodier at night, and the harmonic shift underscores that transition so the player feels the change as much as sees it.

Are there elements to the audio/music that are diegetic/non-diegetic?

There are elements in our audio/music that are diegetic/non-diegetic in our game scene. In terms of diegetic sounds, we have birds chirping in the background, the location-based

waterfall noise, the player's footsteps, the jump sound, and the death sound, which exist in this world but also double as player feedback confirming actions and signaling that something has occurred. The chirping and waterfall in particular populate the environment, breathing life into the background audio to split up the music. The music on the other hand is non-diegetic, and is played as a generative loop. The OSC reacts to when the day/night switch happens, and when it occurs it swaps the audio played, it is non-diegetic as it's not really part of the immersion or environment, but mostly for the player themselves.

What and how do certain musical/audio features (such as tempo, effects processing, waveform design, timbre, melody/harmony, etc...) contribute to the sound design?

Some features we had in our music was having an additive waveform (odd harmonics) to produce a hollow, slightly retro/8-bit timbre rather than a pure sine or a harsh saw. The short plucky envelope makes notes feel percussive and music-box like. The music tempo is set to 400MS, setting the energy and tying it to game state. Certain sounds such as picking up a coin and the player dying utilize several audio layers mixed together, particularly sine waves in tandem with noise layers to create a more natural, consistent sound font.

Is there immersion present in the sound design?

Immersion is supported by several elements in our game: the music and bird chirps, which are tied to the time of day, the waterfall, which is tied to the player's current location in the level, and the coin pickup and death sounds, which are tied to the player's current coin count. The music is immersive precisely because it never repeats exactly, avoiding the "loop fatigue" that breaks immersion during long play sessions, while the OSC-driven voice switching keeps the audio tethered to events in the world so it feels reactive rather than applied on top. The

waterfall anchors the player in a consistent physical space denoted by the audio and the visuals. Additionally, the player's footsteps, jump sounds, and background bird chirps utilize randomness to give the effect of each sound feeling slightly different while sounds such as picking up a coin adjust their pitch based on the player's coin count. Together, these layers create a soundscape that feels less like something laid over the scene and more like a genuine part of the world the player is exploring.

Group Contributions:

Andrew Degan

- Unity scene mock up
- Bug fixes and sorting out Unity/OSC interactions
- Sound Effects
 - Bird SFX
 - Waterfall SFX
 - Death SFX
 - Footstep SFX
 - Jump SFX
 - Coin Pickup SFX
- Making the Master PD file, including taking signals from unity

Ben Nguyen

I joined the group and we worked off of Andrew's midterm project. I made small adjustments to the unity scene and brought in music from my old midterm project with some adjustments to make it more generative, whilst also importing it into Andrew's master PD file for integration into Unity.

- Added a day/night cycle
- Composed the background soundtrack and added it into Unity